λ_L (nm)	τ_L (FWHM)	Laser mode	NA	Energy range	$\bar{U}_{avg}$ (kJ/cm^3)	$\bar{P}_{avg}$ (kbar)	$\bar{T}_{avg}$ (K)	$\bar{\rho}_{c,avg}$ (10^{21} cm^{-3})	ρ_{crit} (10^{21} cm^{-3})
1040	350 fs	ml	0.8	(3-20)×E_{thB}	11.8	42	2850	4.96	1.03
520	306 fs	ml	0.8	(3-30)×E_{thB}	41.3	116	10090	12.1	4.12
347	280 fs	ml	0.8	(3-30)×E_{thB}	17.0	57	4090	3.45	9.28
355	0.56 ns	slm	0.3	(1.5-2.5)×E_{thBPL}	32.5	95	7880		
355	0.56 ns	slm	0.5	(1.5-5.0)×E_{thBPL}	39.2	111	9560		
355	0.56 ns	slm	0.8	(1.5-10)×E_{thBPL}	24.9	77	6000		
1064	1.02 ns	slm	0.8	(1.5-7.0)×E_{thBPL}	35.8	103	8710		
532	0.95 ns	slm	0.8	(1.5-14)×E_{thBPL}	38.3	109	9340		
355	0.93 ns	slm	0.8	(1.5-12)×E_{thBPL}	36.4	105	8860		
1064	11.2 ns	mlm	0.8	(1.5-10)×E_{thBPL}	12.0	43	2900		
1064	11.2 ns	slm	0.8	(1.5-7.0)×E_{thBPL}	23.6	74	5690		
532	8.8 ns	mlm	0.8	(5.0-250)×E_{thB}	8.5	32	2090		
532	8.8 ns	slm	0.8	(1.5-7.0)×E_{thBPL}	12.5	44	3020		
355	6.8 ns	mlm	0.8	(20-100)×E_{thB}	12.5	44	3020		
355	6.8 ns	slm	0.8	(1.5-2.3)×E_{thBPL}	9.7	36	2360		

Table 3 Mean values of the average energy density, $\bar{U}_{avg}$, of luminescent plasmas for different pulse durations, wavelengths, and *NA*s, together with the corresponding values of plasma pressure, $\bar{P}_{avg}$, and temperature, $\bar{T}_{avg}$. For fs breakdown, also the free-electron density $\rho_{c,avg}$ corresponding to $\bar{U}_{avg}$ is listed, together with the critical free-electron density ρ_{crit}. The maximum number density of electrons that can be ionized in single ionization of the $1b_1$ valence band level is 6.68×10^{22} cm^{-3} [113].

Peak values for plasma energy density, pressure, and temperature are, of course, larger than the average values. The difference can be very pronounced for ns plasmas, which exhibit an inhomogeneous substructure (Figs. 7 and 8). It must be considered when average plasma temperatures derived from $\bar{U}_{avg}$ are compared to spectroscopically determined values that are closer to peak temperatures (see section 5.4).

4.5. Energy partitioning

For laser surgery and material processing, it is of interest to know, which fraction of the incident energy is absorbed by the plasma. However, close to the bubble threshold, E_{abs} cannot well be assessed through transmission measurements because scattering is here stronger than absorption (section 3.5).

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Alternatively, bubble size and energy can be used to characterize energy deposition. This approach is convenient but provides only a lower estimate for the efficiency of energy deposition and its transformation into mechanical energy because the shock wave energy is not considered. The conversion efficiency E_B/E_L as a function of incident laser energy is shown in Fig. 10. Close to threshold, E_B/E_L is very small for fs pulses (< 0.001%) and even smaller for ns pulses (< 0.0001%). In the BPL region well above threshold, it converges to a value E_B/E_L ≈20% for all laser parameters.

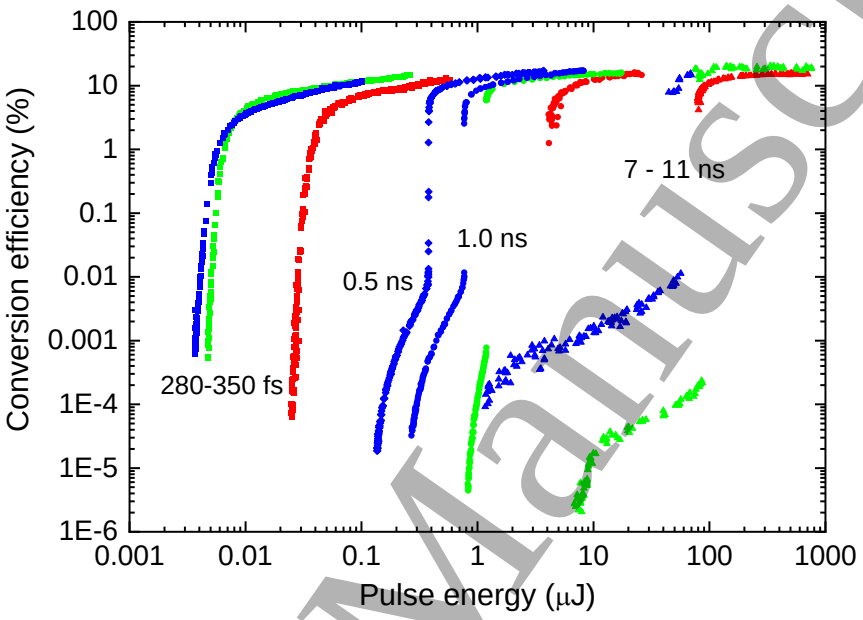

Figure 10 Conversion of laser energy E_L into bubble energy E_B for data of Fig. 5c.

Let us now look at energy partitioning in the luminescent plasma regime well above the bubble threshold, where $\bar{U}_{avg}$ can be determined as shown in Fig. 9. For this regime, we explored the dependence of energy partitioning on plasma energy density based on the data listed in Table 3. Figures 11(a)-(c) show the dependence of conversion ratios E_V/E_{abs}, E_B/E_{abs} and E_{SW}/E_{abs} on $\bar{U}_{avg}$. In Fig. 11(d), averaged values from these data are combined in one graph to present the complete energy balance as a function of $\bar{U}_{avg}$. With increasing plasma energy density, the energy fraction required for vaporization of the liquid in the plasma volume drops since the absolute value of the specific vaporization energy is constant. Therefore, an ever-larger fraction is transformed into mechanical energy, most of it into shock wave energy. Interestingly, the conversion into bubble energy (E_B/E_{abs}) remains approximately constant between 15% and 20% when $\bar{U}_{avg}$ increases from 8.5 kJcm^{-3} to 41 kJcm^{-3}. However, the fraction going into shock wave energy increases from ≈ 55% to ≈ 75%, while the vaporization energy drops from ≈ 30% to little more than 5%.

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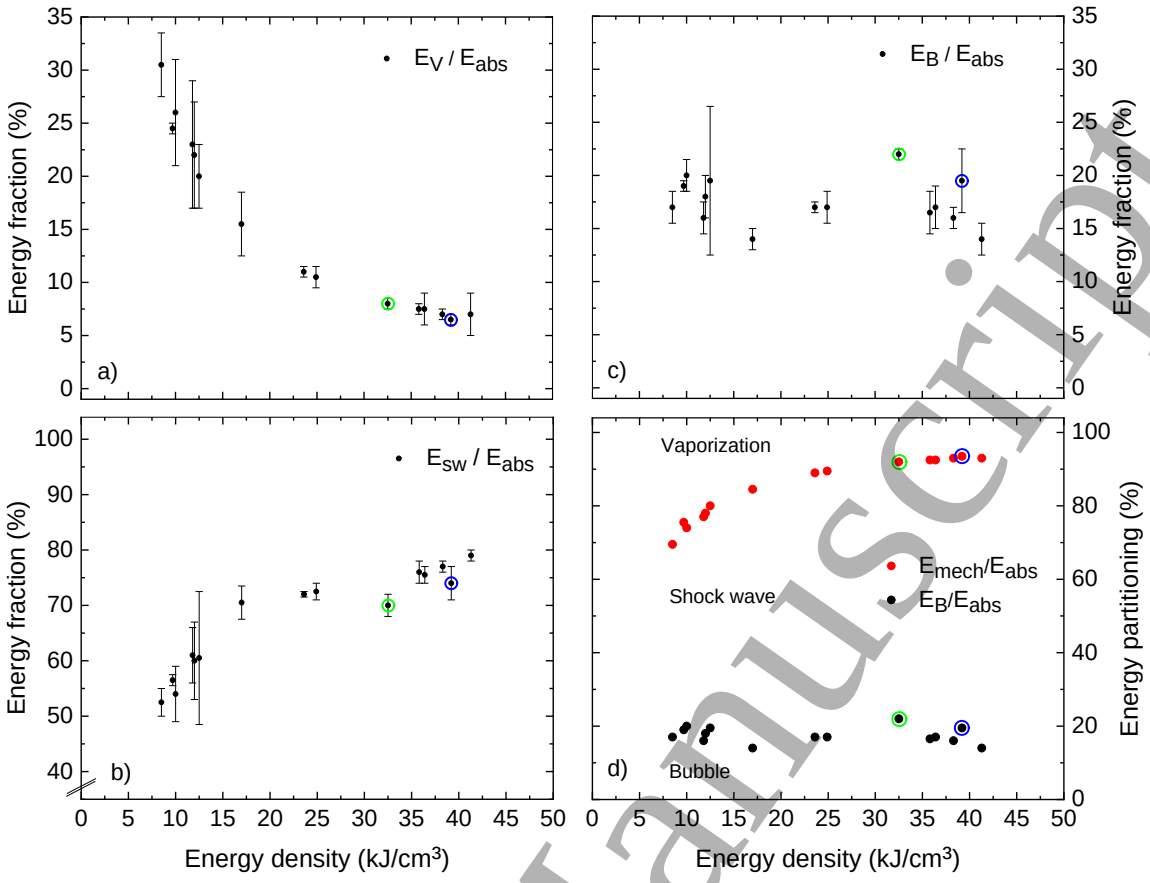

Figure 11 Energy partitioning of absorbed laser energy as a function of average plasma energy density compiled from the results obtained with all laser systems. The individual graphs show the fractions going (a) into vaporization of the liquid within the plasma volume, (b) into shock wave emission, and (c) into bubble formation. All data are mean values averaged over the investigated range of laser pulse energies. The bars denote the variations of E_V/E_{abs}, E_B/E_{abs} and E_{SW}/E_{abs} around the average values. The complete energy balance is shown in (d), where the fractions going into bubble energy, E_B/E_{abs}, and mechanical energy, $(E_{SW} + E_B)/E_{abs}$ are plotted as a function of energy density. Here, $(E_{mech} - E_B)/E_{abs}$ stands for the part going into shock wave energy and $(E_{abs} - E_{mech})/E_{abs}$ represents the fraction needed for vaporization. Most data refer to $NA = 0.8$; values for $NA = 0.3$ and $NA = 0.5$ are marked with green and blue circles, respectively.

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